

Universidad Carlos III de Madrid

Aerospace Engineering

Computer Laboratory Report 2

Particle on Oscillating Loop

Mechanics Applied to Aerospace Engineering

Laboratory Group LA

Authors

Aimar Álvarez Iglesias

Héctor Gonzalez Rodriguez

Santiago Hernández Bejarano

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1 Introduction

In this laboratory the motion of a heavy particle P of mass m that slides without friction on a circular wire of centre C and radius R is studied. The wire lies on a vertical plane Oxy, and this plane turns about the vertical axis Oy with the prescribed angular velocity $\boldsymbol{\Omega}(t) = A\omega \sin(\omega t)\, \boldsymbol{j}$, where A and ω are known positive constants. The position of P on the wire is given by the angle φ between the axis Ox and the vector $\overrightarrow{CP}$ [1].

The aim is to write the equation of motion of P in the rotating frame, integrate it with `ode45`, and use the results to explain the forces of the wire, the energy, the trajectory seen from a fixed frame, when P goes over the top of a fixed loop, and how good the linearized equation is.

Section 2 derives the model and explains the numerical setup. Section 3 discusses the result of each simulation next to its figure, Section 4 gathers the points that apply to the whole model, and Section 5 sums up.

2 Methodology

2.1 Kinematics and degree of freedom

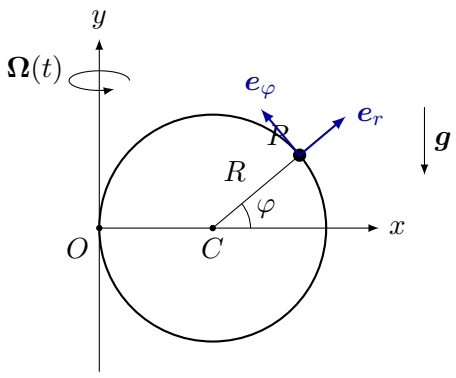


Figure 1: Geometry of the problem. The plane Oxy turns about Oy; $\boldsymbol{k} = \boldsymbol{i} \times \boldsymbol{j}$ points out of the page. The polar basis $\{\boldsymbol{e}_r, \boldsymbol{e}_\varphi\}$ is attached to P.

Two reference frames are used. $Oxyz$ moves with the plane of the wire: $\boldsymbol{i}$ lies along the horizontal diameter, $\boldsymbol{j}$ points upwards along the axis and $\boldsymbol{k} = \boldsymbol{i} \times \boldsymbol{j}$ is normal to the plane. $Ox_1y_1z_1$ is fixed and coincides with $Oxyz$ at $t = 0$. From the figure, the axis Oy is tangent to the wire at O, so $\overrightarrow{OC} = R\,\boldsymbol{i}$ (Figure 1).

Degree of freedom. A free particle has three coordinates. The wire adds two constraints: P must stay in the plane Oxy and at a distance R from C. This leaves $3 - 2 = 1$ degree of freedom, parametrised by φ. The plane turns with a known law, so the position in the fixed frame depends on φ and also explicitly on t.

Motion of the plane. The angle θ turned by the plane satisfies $\dot\theta = \Omega$ with $\theta(0) = 0$. With the change $u = \omega\tau$, $\mathrm{d}u = \omega\,\mathrm{d}\tau$, and Barrow's rule,

$$\theta(t) = \int_0^t A\omega \sin(\omega\tau)\,\mathrm{d}\tau = A\int_0^{\omega t} \sin u\,\mathrm{d}u = A\big[-\cos u\big]_0^{\omega t} = A(1 - \cos\omega t). \tag{1}$$

So the plane swings between $\theta = 0$ and $\theta = 2A$; with $A = \pi$ it makes a full turn in $T/2$ and comes back, where $T = 2\pi/\omega$. A rotation of angle θ about the vertical axis relates both bases: the columns of

$$[{}_1R_0] = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \tag{2}$$

are the components of $\boldsymbol{i}$, $\boldsymbol{j}$, $\boldsymbol{k}$ in $\{\boldsymbol{i}_1, \boldsymbol{j}_1, \boldsymbol{k}_1\}$. The columns are orthonormal, so $[{}_1R_0][{}_1R_0]^T = I$, and $\det[{}_1R_0] = \cos^2\theta + \sin^2\theta = 1$. As a check, $\boldsymbol{i} = \cos\theta\,\boldsymbol{i}_1 - \sin\theta\,\boldsymbol{k}_1$ gives $\mathrm{d}\boldsymbol{i}/\mathrm{d}t = -\dot\theta(\sin\theta\,\boldsymbol{i}_1 + \cos\theta\,\boldsymbol{k}_1) = -\Omega\,\boldsymbol{k} = \boldsymbol{\Omega} \times \boldsymbol{i}$.

Relative motion. Let $\boldsymbol{e}_r = \cos\varphi\,\boldsymbol{i} + \sin\varphi\,\boldsymbol{j}$ and $\boldsymbol{e}_\varphi = -\sin\varphi\,\boldsymbol{i} + \cos\varphi\,\boldsymbol{j}$. The position of P is

$$\boldsymbol{r} = \overrightarrow{OC} + \overrightarrow{CP} = R(1 + \cos\varphi)\,\boldsymbol{i} + R\sin\varphi\,\boldsymbol{j}, \qquad x \equiv R(1 + \cos\varphi), \tag{3}$$

where x is the distance from P to the axis. Differentiating each component in $Oxyz$ with the chain rule, $\frac{\mathrm{d}}{\mathrm{d}t}[R(1 + \cos\varphi)] = -R\dot\varphi\sin\varphi$ and $\frac{\mathrm{d}}{\mathrm{d}t}[R\sin\varphi] = R\dot\varphi\cos\varphi$, so

$$\boldsymbol{v}_{\mathrm{rel}} = -R\dot\varphi\sin\varphi\,\boldsymbol{i} + R\dot\varphi\cos\varphi\,\boldsymbol{j} = R\dot\varphi\,\boldsymbol{e}_\varphi. \tag{4}$$

With the product rule, $\frac{\mathrm{d}}{\mathrm{d}t}(-R\dot\varphi\sin\varphi) = -R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi)$ and $\frac{\mathrm{d}}{\mathrm{d}t}(R\dot\varphi\cos\varphi) = R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)$:

$$\boldsymbol{a}_{\mathrm{rel}} = -R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi)\,\boldsymbol{i} + R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)\,\boldsymbol{j} = R\ddot\varphi\,\boldsymbol{e}_\varphi - R\dot\varphi^2\,\boldsymbol{e}_r. \tag{5}$$

Absolute motion. O is on the axis and does not move, so $\boldsymbol{v} = \boldsymbol{v}_{\mathrm{rel}} + \boldsymbol{\Omega} \times \boldsymbol{r}$ and $\boldsymbol{a} = \boldsymbol{a}_{\mathrm{rel}} + \dot{\boldsymbol{\Omega}} \times \boldsymbol{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r}) + 2\boldsymbol{\Omega} \times \boldsymbol{v}_{\mathrm{rel}}$, with $\Omega = A\omega\sin\omega t$ and $\dot\Omega = A\omega^2\cos\omega t$. Using $\boldsymbol{j} \times \boldsymbol{i} = -\boldsymbol{k}$, $\boldsymbol{j} \times \boldsymbol{j} = \boldsymbol{0}$ and $\boldsymbol{j} \times \boldsymbol{k} = \boldsymbol{i}$, the four products are

$$\boldsymbol{\Omega} \times \boldsymbol{r} = \Omega\,x\,(\boldsymbol{j} \times \boldsymbol{i}) + \Omega R\sin\varphi\,(\boldsymbol{j} \times \boldsymbol{j}) = -\Omega x\,\boldsymbol{k}, \qquad \dot{\boldsymbol{\Omega}} \times \boldsymbol{r} = -\dot\Omega x\,\boldsymbol{k}, \tag{6}$$

$$\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r}) = \Omega\,\boldsymbol{j} \times (-\Omega x\,\boldsymbol{k}) = -\Omega^2 x\,(\boldsymbol{j} \times \boldsymbol{k}) = -\Omega^2 x\,\boldsymbol{i}, \tag{7}$$

$$2\boldsymbol{\Omega} \times \boldsymbol{v}_{\mathrm{rel}} = 2\Omega R\dot\varphi\big[-\sin\varphi\,(\boldsymbol{j} \times \boldsymbol{i}) + \cos\varphi\,(\boldsymbol{j} \times \boldsymbol{j})\big] = 2\Omega R\dot\varphi\sin\varphi\,\boldsymbol{k}. \tag{8}$$

Adding them,

$$\boldsymbol{v} = R\dot\varphi\,\boldsymbol{e}_\varphi - \Omega x\,\boldsymbol{k}, \tag{9}$$

$$\boldsymbol{a} = \big[-R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi) - \Omega^2 x\big]\boldsymbol{i} + R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)\,\boldsymbol{j} + \big[-\dot\Omega x + 2\Omega R\dot\varphi\sin\varphi\big]\boldsymbol{k}. \tag{10}$$

Since $\boldsymbol{e}_\varphi \perp \boldsymbol{k}$, the speed is $|\boldsymbol{v}|^2 = R^2\dot\varphi^2 + \Omega^2 x^2$.

2.2 Forces and equation of motion

Two real forces act on P: its weight and the reaction of the wire. The wire is smooth, so the reaction has no component along the tangent $\boldsymbol{e}_\varphi$:

$$\boldsymbol{W} = -mg\,\boldsymbol{j} = -mg\sin\varphi\,\boldsymbol{e}_r - mg\cos\varphi\,\boldsymbol{e}_\varphi, \qquad \boldsymbol{N} = N_r\,\boldsymbol{e}_r + N_z\,\boldsymbol{k}, \tag{11}$$

where $\boldsymbol{j} = \sin\varphi\,\boldsymbol{e}_r + \cos\varphi\,\boldsymbol{e}_\varphi$ was used. In the rotating frame the inertia forces have to be added. From Equations (6) to (8),

$$\boldsymbol{F}_{\text{cf}} = -m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) = m\Omega^2 x\,\boldsymbol{i} = mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\big[\cos\varphi\,\boldsymbol{e}_r - \sin\varphi\,\boldsymbol{e}_\varphi\big], \tag{12}$$

$$\boldsymbol{F}_{\text{Eul}} = -m\dot{\boldsymbol{\Omega}}\times\boldsymbol{r} = m\dot{\Omega}x\,\boldsymbol{k} = mA\omega^2 R\cos(\omega t)(1+\cos\varphi)\,\boldsymbol{k}, \tag{13}$$

$$\boldsymbol{F}_{\text{Cor}} = -2m\boldsymbol{\Omega}\times\boldsymbol{v}_{\text{rel}} = -2m\Omega R\dot\varphi\sin\varphi\,\boldsymbol{k} = -2mA\omega R\dot\varphi\sin(\omega t)\sin\varphi\,\boldsymbol{k}, \tag{14}$$

with $\boldsymbol{i} = \cos\varphi\,\boldsymbol{e}_r - \sin\varphi\,\boldsymbol{e}_\varphi$. Newton's second law in $Oxyz$ reads $m\boldsymbol{a}_{\text{rel}} = \boldsymbol{W}+\boldsymbol{N}+\boldsymbol{F}_{\text{cf}}+\boldsymbol{F}_{\text{Eul}}+\boldsymbol{F}_{\text{Cor}}$. The tangent $\boldsymbol{e}_\varphi$ is the only direction free of the unknown reaction, and the Euler and Coriolis forces point along $\boldsymbol{k}$, so projecting on $\boldsymbol{e}_\varphi$ gives $mR\ddot\varphi = -mg\cos\varphi - m\Omega^2 x\sin\varphi$, that is

$$\ddot\varphi = -\frac{g}{R}\cos\varphi - A^2\omega^2\sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi. \tag{15}$$

The other two projections give the reaction. Along $\boldsymbol{e}_r$, $-mR\dot\varphi^2 = N_r - mg\sin\varphi + m\Omega^2 x\cos\varphi$, and along $\boldsymbol{k}$, $0 = N_z + m\dot\Omega x - 2m\Omega R\dot\varphi\sin\varphi$:

$$N_r = -mR\dot\varphi^2 + mg\sin\varphi - mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\cos\varphi, \tag{16}$$

$$N_z = mA\omega R\big[2\dot\varphi\sin(\omega t)\sin\varphi - \omega\cos(\omega t)(1+\cos\varphi)\big]. \tag{17}$$

The meaning of each term in Equation (15) is the following:

- If the plane does not turn ($A = 0$), the equation is the one of a pendulum, $\ddot\varphi = -(g/R)\cos\varphi$, with φ measured from the horizontal. The bottom, $\varphi = -\pi/2$, is a stable equilibrium and the top, $\varphi = \pi/2$, an unstable one.
- Only the centrifugal force enters the equation, because it is the only inertia force in the plane of the wire. The Euler and Coriolis forces are normal to the plane and only load the wire through N_z.
- The centrifugal term is at most $(A\omega)^2/(g/R) = \pi^2\cdot 0.01/9.81 \approx 1\%$ of the gravity term for the data of item 5, so the rotation is a weak perturbation of the pendulum.
- The wire holds P on both sides, so $\boldsymbol{N}$ can take any sign. Unlike the string of Lab 1, there is no condition that stops the model.

2.3 Mechanical energy

With the absolute velocity of Equation (9), the kinetic and gravitational potential energies are

$$E_k = \tfrac{1}{2}m(R^2\dot\varphi^2 + \Omega^2 x^2), \qquad E_p = mgR\sin\varphi, \qquad E = E_k + E_p. \tag{18}$$

In the fixed frame the weight is included in E, so $\mathrm{d}E/\mathrm{d}t$ is the power of the reaction. Using Equations (9) and (11) and $\boldsymbol{e}_r\cdot\boldsymbol{e}_\varphi = 0$,

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \boldsymbol{N}\cdot\boldsymbol{v} = (N_r\,\boldsymbol{e}_r + N_z\,\boldsymbol{k})\cdot(R\dot\varphi\,\boldsymbol{e}_\varphi - \Omega x\,\boldsymbol{k}) = -\Omega\,x\,N_z. \tag{19}$$

The wire moves with the plane and pushes P along $\boldsymbol{k}$, so it does work whenever $\Omega \neq 0$: E is conserved only for $A = 0$.

2.4 Fixed loop: condition to go over the top

For $A = 0$, multiplying Equation (15) by $\dot\varphi$ gives $\dot\varphi\ddot\varphi + (g/R)\cos\varphi\,\dot\varphi = 0$. By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}t}(\frac{1}{2}\dot\varphi^2) = \dot\varphi\ddot\varphi$ and $\frac{\mathrm{d}}{\mathrm{d}t}\sin\varphi = \cos\varphi\,\dot\varphi$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big[\tfrac{1}{2}\dot\varphi^2 + \frac{g}{R}\sin\varphi\Big] = 0 \quad\Rightarrow\quad \tfrac{1}{2}\dot\varphi^2 + \frac{g}{R}\sin\varphi = \tfrac{1}{2}\dot\varphi_0^2 + \frac{g}{R}\sin\varphi_0, \tag{20}$$

which is $E/(mR^2) = \text{constant}$. The potential energy is largest at the top, $\varphi = \pi/2$, where $E_p = mgR$. If P reaches the top with $\dot\varphi \neq 0$ it goes over it and φ keeps growing; otherwise it stops before and comes back. Setting $\dot\varphi^2 > 0$ at $\varphi = \pi/2$ in Equation (20) gives the criterion

$$|\dot\varphi_0| > \dot\varphi_c = \sqrt{\frac{2g}{R}(1 - \sin\varphi_0)}. \tag{21}$$

The equality defines the separatrix $\dot\varphi = \pm\sqrt{(2g/R)(1-\sin\varphi)}$, along which P approaches the top but needs an infinite time to reach it. Below it, φ oscillates between the two angles where $\dot\varphi = 0$, that is, $\sin\varphi = \sin\varphi_0 + R\dot\varphi_0^2/(2g)$. Table 1 applies the criterion to the nine initial conditions of item 7.

Table 1: Critical rate of Equation (21) ($g/R = 9.81\,\mathrm{s}^{-2}$) and predicted motion for the nine initial conditions.

φ_0 (rad)	$\dot\varphi_c$ (rad/s)	$\dot\varphi_0 = 0$	$\dot\varphi_0 = 1.5$ rad/s	$\dot\varphi_0 = 3$ rad/s
0	4.429	oscillates	oscillates	oscillates
$\pi/4$	2.397	oscillates	oscillates	goes over the top
$\pi/2$	0	rest at the top (unstable)	goes over the top	goes over the top

2.5 Linearized equation

Now $A^2 \ll g/(R\omega^2)$ and $\varphi = \delta - \pi/2$ with $|\delta| \ll 1$. Then $\cos\varphi = \cos(\delta - \pi/2) = \sin\delta \approx \delta$, $\sin\varphi = -\cos\delta \approx -1$, $1 + \cos\varphi \approx 1 + \delta$ and $\ddot\varphi = \ddot\delta$. Substituting in Equation (15),

$$\ddot\delta = -\frac{g}{R}\,\delta + \Omega^2(1+\delta) = -\frac{g}{R}\,\delta + \Omega^2 + \Omega^2\delta. \tag{22}$$

The hypothesis $A^2 \ll g/(R\omega^2)$ means $\Omega^2 \le A^2\omega^2 \ll g/R$, so the term $\Omega^2\delta$ is small next to $(g/R)\delta$ and is dropped. With $\sin^2(\omega t) = \frac{1}{2}[1 - \cos(2\omega t)]$,

$$\ddot\delta + \omega_n^2\,\delta = \Omega^2 = F_0[1 - \cos(2\omega t)], \qquad \omega_n = \sqrt{g/R}, \qquad F_0 = \tfrac{1}{2}A^2\omega^2. \tag{23}$$

This is a forced oscillator without damping: natural frequency ω_n and forcing at 2ω [2]. The statement gives no initial conditions for item 8, and they are not given in the problem sheet either [3]. Here P starts at rest at the bottom, $\delta(0) = 0$ and $\dot\delta(0) = 0$, so all its motion comes from the loop. The particular solution is a constant F_0/ω_n^2 plus $C\cos(2\omega t)$; substituting the second one, $(\omega_n^2 - 4\omega^2)C = -F_0$. Adding the free solution $a\cos\omega_n t + b\sin\omega_n t$ and imposing the initial conditions ($\dot\delta(0) = b\,\omega_n = 0$),

$$\delta(t) = \frac{F_0}{\omega_n^2} - \frac{F_0\cos(2\omega t)}{\omega_n^2 - 4\omega^2} + \Big(\frac{F_0}{\omega_n^2 - 4\omega^2} - \frac{F_0}{\omega_n^2}\Big)\cos(\omega_n t), \qquad 2\omega \neq \omega_n. \tag{24}$$

So P oscillates about $\delta_s = F_0/\omega_n^2 = A^2R\omega^2/(2g)$, slightly outwards from the bottom. When $2\omega = \omega_n$ the denominator vanishes. This is resonance, and the particular solution changes to $-\frac{F_0}{2\omega_n}t\sin(\omega_n t)$, which satisfies $\ddot\delta + \omega_n^2\delta = -F_0\cos(\omega_n t)$. With the same initial conditions,

$$\delta(t) = \frac{F_0}{\omega_n^2}(1 - \cos\omega_n t) - \frac{F_0}{2\omega_n}\,t\sin(\omega_n t), \qquad 2\omega = \omega_n, \tag{25}$$

whose amplitude grows linearly with time. Sooner or later $\delta \ll 1$ fails, and the linearization with it.

2.6 Numerical method

Equation (15) is a second order nonlinear equation with time-dependent coefficients. To integrate it with `ode45` [4] it is written as a first order system for the state vector $\boldsymbol{X} = [\varphi, \dot\varphi]^T$:

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \begin{pmatrix} X_2 \\ -\dfrac{g}{R}\cos X_1 - \Omega(t)^2\,(1+\cos X_1)\sin X_1 \end{pmatrix}, \qquad \boldsymbol{X}(0) = \begin{pmatrix}\varphi_0 \\ \dot\varphi_0\end{pmatrix}, \qquad \Omega(t) = A\omega\sin(\omega t). \tag{26}$$

This system is the MATLAB function `diffeq(t,X,g,w,R,A)`, and Equation (23) is `diffeqlin` with the state $[\delta, \dot{\delta}]^T$. The tolerances are 10^{-8} (relative) and 10^{-10} (absolute). In item 5 the solution is stored at 10 000 instants in $[0, 3T]$, about 125 points for each oscillation of φ, well above the sampling needed to draw it [1]. The forces and energies are then computed from the stored φ and $\dot{\varphi}$ with Equations (12) to (14) and (16) to (18). To check Equation (19) without adding a state, the work of the wire is computed afterwards from the power $P_i = -\Omega_i x_i N_{z,i}$ with the trapezoidal rule,

$$W_n = \sum_{i=1}^{n-1} \tfrac{1}{2}\,(P_i + P_{i+1})(t_{i+1} - t_i), \tag{27}$$

and compared with $E - E_0$. The position in the fixed frame follows from Equations (1) and (2): $x_1 = x\cos\theta$, $y_1 = R\sin\varphi$, $z_1 = -x\sin\theta$. In all cases $g = 9.81\,\mathrm{m/s^2}$ and $m = 1\,\mathrm{kg}$; the three simulations are listed in Table 2.

Table 2: Simulations. In item 8, $\omega_n = \sqrt{g/R} = 31.32\,\mathrm{rad/s}$.

Item	Case	A (rad)	R (m)	ω (rad/s)	Initial conditions and time
5, 6	oscillating loop	π	1	0.1	$\varphi_0 = 0$, $\dot{\varphi}_0 = 0$; $t \in [0, 3T]$, $T = 62.83\,\mathrm{s}$
7	fixed loop	0	1	–	Table 1, and $\varphi_0 = \pi/4$, $\dot{\varphi}_0 = (1 \pm 0.01)\,\dot{\varphi}_c$; $t \in [0, 4]$ s
8	linearization	0.1	0.01	ω_n, $\omega_n/2$, $\omega_n/4$	$\delta_0 = 0$, $\dot{\delta}_0 = 0$; $t \in [0, 10]$ s

3 Results

3.1 Oscillating loop: angle, reaction and inertia forces

At $t = 0$ the loop is at rest ($\Omega(0) = 0$) and Equation (15) gives $\ddot{\varphi}(0) = -g/R = -9.81\,\mathrm{rad/s^2}$, the value printed by the code. P falls from the end of the horizontal diameter and swings between $\varphi = 0$ and $\varphi = -\pi$ (Figure 2a,b). At $\varphi = -\pi$ it is at O, on the axis. The period is 2.37 s, longer than $2\pi\sqrt{R/g} = 2.006\,\mathrm{s}$ because the amplitude about the bottom is 90° and the small-oscillation result does not apply. The amplitude does not change over the three periods of the loop: the centrifugal term is only 1% of gravity. The largest angle is $\varphi = 0.0118\,\mathrm{rad}$, so the energy given by the wire lifts P about 1 cm above its starting height.

The radial reaction (Figure 2c) goes from 0 at the turning points to $-29.43\,\mathrm{N}$ at the bottom. Its value is $-3mg$: at the bottom Equation (20) gives $\dot{\varphi}^2 = 2g/R$, and Equation (16) gives $N_r = -2mg - mg$. The sign is negative because the wire has to pull P towards C to give it the centripetal acceleration. The normal reaction (Figure 2d) reaches $\pm 2.78\,\mathrm{N}$. Almost all of it balances the Coriolis force (Figure 2f), which is $2mA\omega R\dot{\varphi}_{\max} = 2\pi(0.1)(4.43) = 2.78\,\mathrm{N}$ at most. It is large because P moves fast along the wire ($\dot{\varphi}$ up to $4.4\,\mathrm{rad/s}$) while the plane turns. Its envelope follows $|\sin\omega t|$ and almost vanishes at $t = 0, T/2, T, \ldots$, when the plane stops to reverse and only the small Euler term is left. The centrifugal force (Figure 2e) is always positive, pointing away from the axis, and peaks at $m(A\omega)^2(2R) = 0.197\,\mathrm{N}$ at $t = T/4, 3T/4, \ldots$, when Ω^2 is largest. The Euler force follows $\cos\omega t$ and stays below 0.063 N, because the loop changes its rate slowly: $|\dot{\Omega}| \le A\omega^2 = 0.031\,\mathrm{rad/s^2}$.

3.2 Energy

E_k and E_p exchange about 9.8 J in every oscillation (Figure 3a), as in a pendulum. Their sum is not constant (Figure 3b): $E - E_0$ moves between $-0.082\,\mathrm{J}$ and $0.313\,\mathrm{J}$, with a fast oscillation at the frequency of φ inside an envelope that follows Ω^2. While $|\Omega|$ grows, the wire pushes P along the direction in which the plane moves and gives it energy, and while $|\Omega|$ decreases it takes the energy back. The loop changes its rate slowly compared with the oscillation ($T = 62.8\,\mathrm{s}$ against 2.37 s), so the energy returns to E_0 every time the plane stops, and after $3T$ the net gain is only $0.0006\,\mathrm{J}$. This also explains why the amplitude of φ in Figure 2a does not drift.

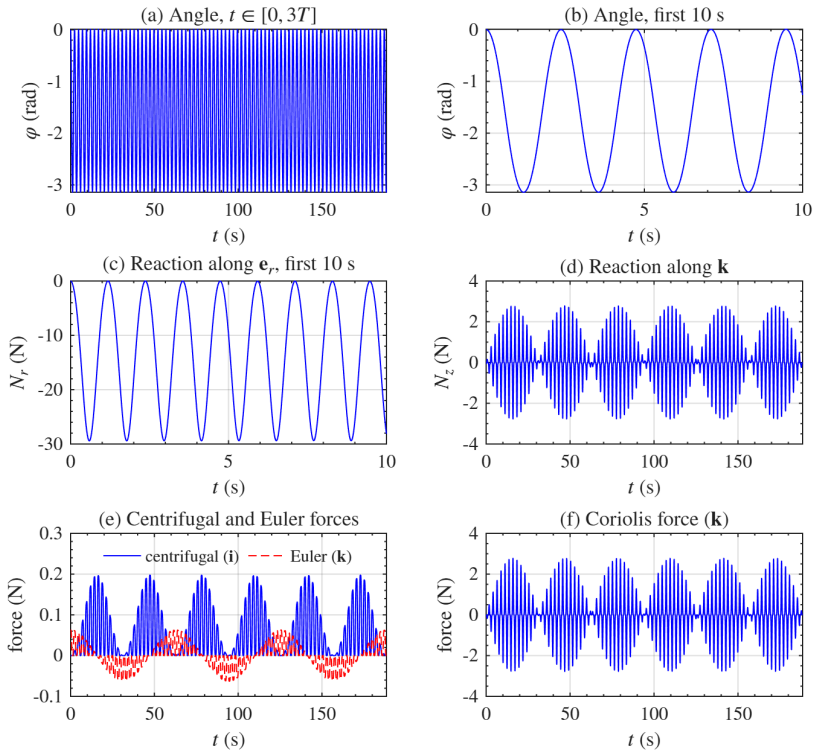


Figure 2: Oscillating loop, item 5. (a) Angle in $[0, 3T]$ and (b) in the first 10 s. (c) Reaction along $\boldsymbol{e}_r$, first 10 s. (d) Reaction along $\boldsymbol{k}$. (e) Centrifugal ($\boldsymbol{i}$) and Euler ($\boldsymbol{k}$) forces. (f) Coriolis force ($\boldsymbol{k}$).

The work of the wire from Equation (27) lies on top of $E - E_0$; the largest difference is 1.5×10^{-4} J, 0.05% of the range of E, and comes from the trapezoidal rule on the stored points. This confirms Equation (19) and the expressions of N_z and E used in the code. Repeating the integration with tolerances 10^{-11} and 10^{-12} changes φ by at most 1.4×10^{-4} rad.

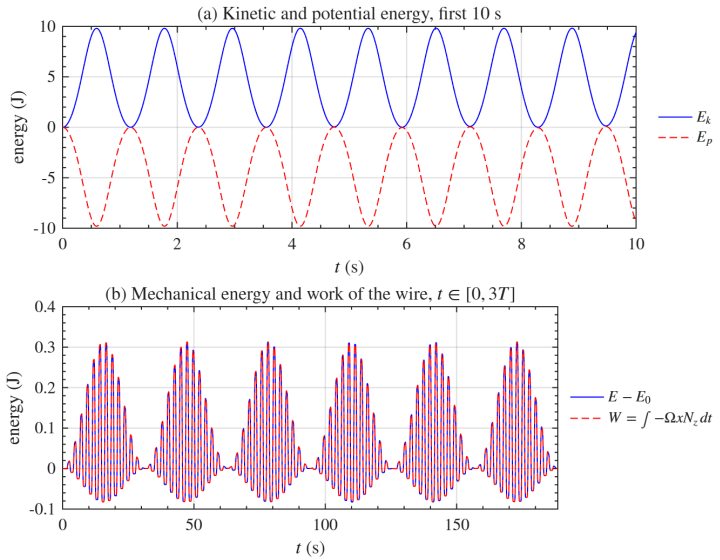


Figure 3: Energy, item 5. (a) Kinetic and potential energy in the first 10 s; the same pattern repeats over the whole interval. (b) Increment of mechanical energy and work of the wire, Equation (27), in $[0, 3T]$.

3.3 Trajectory in the fixed frame

P always stays on the surface obtained by turning the circle about Oy, a torus whose hole has closed because the circle touches the axis at O (Figure 4a). Seen from above (Figure 4b), each oscillation of φ draws one petal: P goes out to $x = 2R$ when $\varphi = 0$ and back to the axis when $\varphi = -\pi$. The petals spread around the axis because the plane turns from $\theta = 0$ to $\theta = 2\pi$ at $t = T/2$ and back to 0 at $t = T$ (Equation (1)). The three periods draw almost the same curve; they are not identical because the period of φ (2.37 s) does not divide T. As a check, $\sqrt{x_1^2 + z_1^2}$ equals x to 2×10^{-16} m, since the rotation in Equation (2) does not change the distance to the axis.

The script also writes an animation of the first period, `trajectory_P.avi`, which is included in the submitted folder. For each frame the wire is drawn turned by $\theta(t_k)$, together with the axis, the trail of P up to t_k and the particle. One frame is taken every 6 stored points, about 20 frames for each oscillation of φ, so the motion is not undersampled; this gives 556 frames, played at 30 frames per second. The guidelines suggest `getframe` [1], but when the code was run without a screen it sometimes returned frames of different sizes, which `VideoWriter` rejects. Each frame is therefore printed to a fixed 560×480 image with `print(gcf,'-RGBImage','-r80')` and added to the video with `writeVideo`.

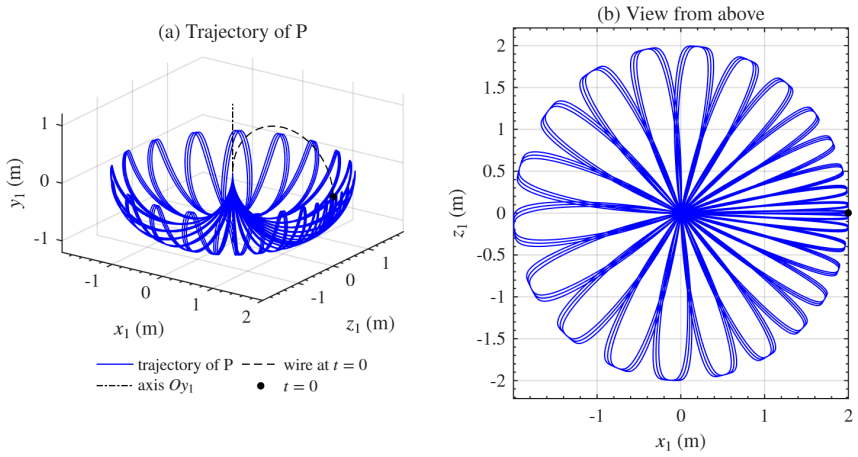


Figure 4: Trajectory of P in the fixed frame $Ox_1y_1z_1$ for $t \in [0, 3T]$. (a) Three-dimensional view with the wire at $t = 0$ and the axis of rotation. (b) View from above.

3.4 Fixed loop: phase portrait

Figure 5a shows the nine cases of Table 1 with the separatrix of Equation (21). Five cases, $(0, 0)$, $(0, 1.5)$, $(0, 3)$, $(\pi/4, 0)$ and $(\pi/4, 1.5)$, give closed curves around the stable equilibrium $(-\pi/2, 0)$: φ oscillates, as predicted. Their periods grow from 2.37 s to 3.36 s as the curve gets closer to the separatrix, because P moves slowly near the top. The cases $(\pi/4, 3)$, $(\pi/2, 1.5)$ and $(\pi/2, 3)$ start above the separatrix and give open curves: φ increases steadily, with the lowest $\dot{\varphi}$ at the top and the highest at the bottom of each turn. The case $(\pi/2, 0)$ is the unstable equilibrium and stays as a point; numerically $\cos(\pi/2) \approx 6 \times 10^{-17}$, too small to move it in 4 s. All nine cases agree with Table 1.

To test the threshold itself, $\varphi_0 = \pi/4$ was integrated 1% below and 1% above $\dot{\varphi}_c$ (Figure 5b). Both curves approach the top and stay close to it for almost one second, which is the slow motion near the separatrix. The first one reaches $\varphi = 1.463$ rad $< \pi/2$ and falls back; the second one goes over the top. The energy of Equation (20) is conserved to better than $2 \times 10^{-7}\, mgR$ in all the cases.

3.5 Linearized and full equation

Figure 6a shows the percentage error $100\,|\delta_{\text{lin}} - \delta_{\text{full}}| / \max|\delta_{\text{full}}|$. It is divided by the amplitude of δ and not by δ itself, because δ crosses zero and the error relative to it would diverge there. The thick lines are the largest error reached up to each time. The `ode45` solution of Equation (23)

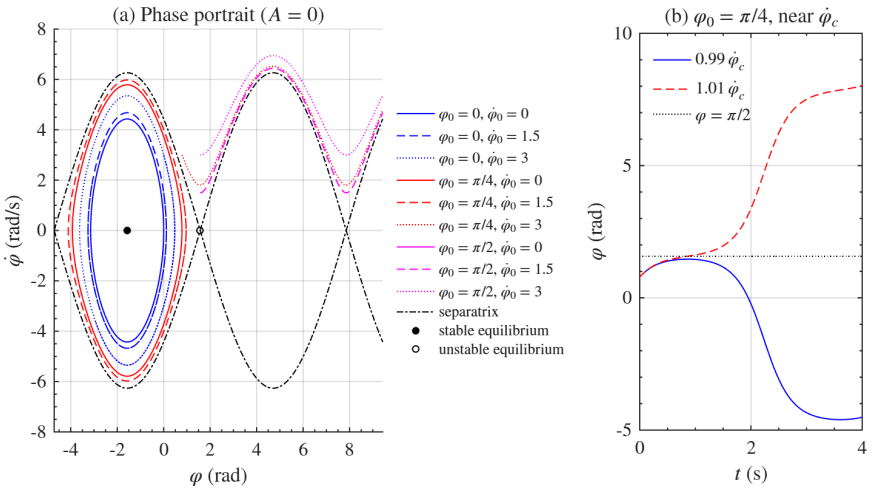


Figure 5: Fixed loop ($A = 0$), item 7. (a) Phase portrait for the nine initial conditions, with the separatrix and the two equilibria. (b) $\varphi_0 = \pi/4$ with $\dot{\varphi}_0$ 1% below and above the critical rate of Equation (21).

matches the closed forms Equations (24) and (25) to 2×10^{-8} rad, so the differences come from the model and not from the integration. Table 3 gives the numbers.

Table 3: Linearized against full equation, $t \in [0, 10]$ s.

ω	$\max \lvert\delta_{\text{full}}\rvert$ (rad)	error at $t = 5$ s	largest error
$\sqrt{g/R}$	0.0134	6.68%	21.4%
$\frac{1}{2}\sqrt{g/R}$	0.196	3.30%	27.8%
$\frac{1}{4}\sqrt{g/R}$	0.0008	0.32%	1.86%

For $\omega = \sqrt{g/R}$, δ stays small but the error grows steadily up to 21%. The reason is the term $\Omega^2\delta$ dropped in Section 2.5: it reduces the stiffness of the full equation from $g/R = 981\,\text{s}^{-2}$ by about $A^2\omega^2/2 = 4.9\,\text{s}^{-2}$ on average, so the full solution oscillates about 0.25% slower and falls behind the linear one. A frequency difference produces a phase lag that grows linearly with time, of about 0.8 rad after 10 s. For $\omega = \frac{1}{2}\sqrt{g/R}$ the forcing frequency 2ω equals ω_n: this is the resonance of Equation (25) [2]. Both solutions grow linearly (Figure 6b) and reach 0.196 rad (11°) after 10 s. The error stays small at first, but as δ grows, $\sin\delta \approx \delta$ stops being accurate and the full pendulum, whose period gets longer with the amplitude, slips out of step with the forcing; the error reaches 27.8%. Without damping, the linear equation predicts unbounded growth, so for this ω the hypothesis $\delta \ll 1$ always fails in the end. For $\omega = \frac{1}{4}\sqrt{g/R}$ the forcing is far from resonance and 16 times weaker than in the first case: δ stays below 8×10^{-4} rad, the dropped term is negligible, and the error stays below 2%. This is the case where the linearization works.

4 Discussion

Some points apply to the whole model and not to a single figure:

- The equation of motion does not depend on m, and the rotation enters only through the centrifugal term. With $(A\omega)^2/(g/R) \approx 1\%$, item 5 is close to a fixed pendulum; the rotation shows mostly in the reaction N_z and in the energy.
- The mechanical energy in the fixed frame is not conserved because the wire moves (Equation (19)). In the rotating frame the reaction, the Euler force and the Coriolis force do no work, since they are normal to $\boldsymbol{v}_{\text{rel}}$, but the centrifugal potential $-\frac{1}{2}m\Omega^2x^2$ depends on time, so $E_k^{\text{rel}} + E_p - \frac{1}{2}m\Omega^2x^2$ is not conserved either. Only for constant Ω, in particular $A = 0$, is there a conserved energy, and the phase portrait of item 7 relies on it.

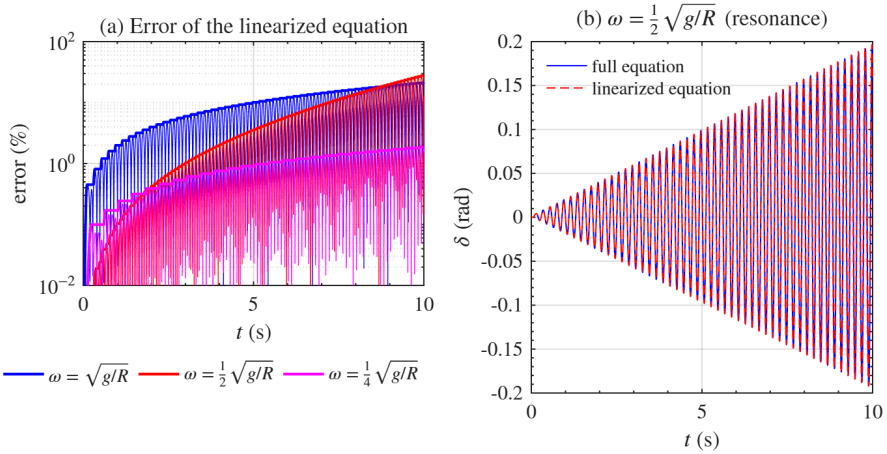


Figure 6: Item 8, $R = 0.01\,\mathrm{m}$, $A = 0.1$. (a) Percentage error of the linearized equation for the three frequencies; thin lines: error, thick lines: largest error up to each time. (b) Full and linearized solutions at resonance.

- The seven checks of the code ($\ddot{\varphi}(0)$, $N_r = -3mg$, $E - E_0 = W$, energy for $A = 0$, distance to the axis, closed form of the linear solution and tolerances) test different parts of it, and all agree with Section 2.
- The results of item 8 depend on the initial conditions, which the statement does not give. With $\delta(0) \neq 0$ a free oscillation of that amplitude would be added and the error would be divided by a larger $\max|\delta|$, but the two mechanisms, the phase lag and the resonance, would be the same.

5 Conclusions

The motion of a particle on a circular wire whose plane oscillates about a vertical axis was analysed in the rotating frame and integrated with `ode45`. The system has one degree of freedom, φ, and its equation contains only gravity and the centrifugal force. The Euler and Coriolis forces act normal to the plane and are balanced by the wire.

For the data of item 5 the rotation is slow and weak, so the particle swings like a pendulum of 90° amplitude about the bottom with a period of 2.37 s. The wire has to supply up to $3mg$ in its plane and up to 2.78 N normal to it, almost all against the Coriolis force. The mechanical energy is not conserved because this normal reaction does work, but the work integrates back to almost zero every time the plane stops. Seen from a fixed frame, the particle draws a rosette of petals on the surface swept by the wire.

With the loop fixed, energy conservation gives the critical rate $\dot{\varphi}_c = \sqrt{(2g/R)(1 - \sin\varphi_0)}$, and the nine initial conditions and the two cases at $\pm 1\%$ of $\dot{\varphi}_c$ follow it. Finally, the linearized equation is a forced oscillator. It is accurate when the forcing is weak and far from resonance, it loses phase when the dropped term $\Omega^2\delta$ is not small enough, and it fails at resonance, $\omega = \frac{1}{2}\sqrt{g/R}$, where the amplitude grows until $\delta \ll 1$ no longer holds.

References

[1] Aerospace Engineering Department (UC3M), *Mechanics Applied to Aerospace Engineering: Laboratory Guidelines*, Version 2026.

[2] Aerospace Engineering Department (UC3M), *Mechanics Applied to Aerospace Engineering: course notes*, Sec. 5.4, “Damped oscillator under a periodic force”.

[3] Aerospace Engineering Department (UC3M), *Mechanics Applied to Aerospace Engineering: particle exercises and problems*, Problem 34.

[4] The MathWorks, Inc., *MATLAB Documentation: ode45, odeset, VideoWriter, print.*